

Q&A preparation

p -adic Variational Quantum Circuits Have No Barren Plateaus

AIQxQIA 2026, Perugia · Thursday 8 October, 11:55–12:15, room B3 · 17 min + 3 min questions

General rule. On questions 2 and 3, *concede first, then narrow the claim*. A precise answer comes across stronger than a defensive one. Never claim more than the paper proves: the theorem is conditional on the splitting hypothesis, controls only the leading digit, and predicts nothing about any existing device.

1. “No hardware exists and you can’t even sample. Isn’t this physically meaningless?”

Why it hurts: This is the first thing an AIQxQIA audience will think. If you sound defensive, the rest of the talk loses value.

Answer (spoken):

“Agreed: no p -adic quantum computer exists, and the model has no Monte Carlo readout, so it predicts nothing about a device. I use it as a **controlled experiment on the complex theory**. I keep the circuit, the observable, the ansatz and the qubit count fixed, and change only the field. The plateau disappears.

So barren plateaus are not caused by unitarity or Hilbert-space dimension alone. The load is carried by two properties of $\mathbb{C}$: **compactness** of the unitary group and the **Archimedean** absolute value. That tells a mitigation where it has to bite. Restricting the ensemble away from Haar-typical, which is what structured ansätze and warm starts do, attacks an ingredient the argument genuinely needs.”

Then add: the method is reusable, since any Haar moment over G_N becomes a count in a finite unitary group (overlaps, expectation values, p -adic kernels). Any application would come through classical p -adic learning on hierarchical data, where the geometry is ultrametric to begin with.

Don’t say: “hardware may come one day.” The paper explicitly says it is on no roadmap. Also do not rank the complex-side mitigations: the paper does not say which one is more effective.

Slides to show: “Why study a model with no hardware?”

2. “Isn’t this trivial, or a plateau in disguise? Your gradient is always at most p^{-1} , and the result is just ‘a random p -adic integer is a unit with probability $1 - 1/p$.’”

Why it hurts: It attacks the content of the result, not its relevance. It also hides a sharper follow-up: “with $p = 3$, a third of the time you fail.”

Answer (spoken):

(a) p^{-1} is a **ceiling, not a plateau**. “ $\exp_p$ only converges when the generator is divisible by p , and because the value group is discrete there is no slack: $\|X\| < p^{-1/(p-1)}$ is *equivalent* to $\|X\| \leq p^{-1}$. So p^{-1} is the largest gradient *any* convergent ansatz can have, and it is a constant: it does not shrink with n . A plateau is something that degrades with the qubit count; this does not.”

(b) **The heuristic predicts the answer; the theorem makes it true and says when it fails.** “Three things had to be proved:

- the natural ensemble does not exist: the full unitary group is non-compact, so there is no Haar probability measure, and one must pass to $G_N = U_N(\mathcal{O})$;
- the random state is uniform on a finite sphere: that is the transitivity lemma;
- the error is controlled *uniformly in n* : that is the counting lemma, with error $10p^{1-r}$.

The heuristic is *false* on the commuting locus $[\widehat{M}, A] \equiv 0 \bmod p$, and the theorem identifies that locus exactly.”

(c) **The $1/p$ failure is mild.** “Failing means the gradient is p^{-2} , one digit lower, not exponentially small. The expected valuation should be $O(1)$: I state that as a conjecture, not as a theorem.”

Don’t say: “the $1/p$ is negligible.” For $p = 3$ it is not. Say it is *bounded and independent of n* , which is the point.

Slides to show: “Surprise 2: the benchmark is p^{-1} ”, “Reading the theorem”, backup “the bad set is a commutator”.

3. “Your theorem assumes the commutator splits over $\mathbb{F}_p$. Isn’t that hypothesis doing all the work? How many circuits actually satisfy it?”

Why it hurts: This is the real weak spot of the paper, and a mathematician will find it. The follow-up is: “your corollary with random U_B needs splitting for almost every U_B . Have you proved that?” You have not.

Answer (spoken):

“That is the main gap, and I say so in the paper. Over a finite field a Hermitian matrix need not diagonalise over the fixed field: I give a 2×2 counterexample over $\mathbb{F}_9$, with eigenvalues $\pm \iota$ outside $\mathbb{F}_3$. So the theorem is **conditional** on splitting, and I do not know what happens outside that class.

Two things I can say.

- The class is not exotic: the most ordinary pair you would write down, Pauli X against a phase rotation, satisfies it with $r = 2^{n-1}$, which is as large as the pigeonhole bound allows.
- The hypothesis is **checkable**: reduce two integer matrices modulo p and look at the characteristic polynomial.

For the random- U_B version I need splitting for almost every U_B , and I have not shown that. Closing the gap needs the classification of **pencils of Hermitian forms** over a finite field, which is the next step. The most useful immediate companion is exact enumeration in $\mathcal{O}/\mathfrak{m}^k$ for small n and p .”

Don’t say: “it holds generically.” That is not proved.

Slides to show: “Limitations”, backup “why the splitting hypothesis is not free”, backup “how large can r be?”.

4. Quick replies for the follow-ups

Q: Does a maximal gradient mean the circuit trains?

No. I remove gradient concentration from the list of obstructions; convergence then depends on conditioning. With a scalar step, p -adic descent contracts only if the Hessian is scalar modulo p . A

matrix preconditioner relaxes that to the Hessian being *invertible* modulo p , which is generic. When it holds, convergence is ultra-fast: $O(\log_p(1/\varepsilon))$ iterations.

Slides to show: “Does a maximal gradient imply trainability?”

Q: *Why G_N and not the group generated by your gates?*

G_N is the smallest natural compact group containing every realisable circuit (Pauli, CNOT, the $\text{SO}(3)_p$ gates, Hadamard when $\sqrt{2} \in K$). A gate-set-specific analysis would give sharper results and is the hardware-aware next step.

Slides to show: “The fix: integral unitaries”.

Q: *Did you verify any of this numerically?*

Not yet, and sampling cannot be used: generalised probabilities lie in $\mathbb{Q}_p$, not in $[0, 1]$. The right check is exact counting over the finite rings $\mathcal{O}/\mathfrak{m}^k$, which I plan to do next.

Slides to show: “One thing that does not carry over: sampling”, “Limitations”.

Q: *What about $p = 2$, or a ramified extension?*

Not covered. Both change the residue field and the norm group, and $p = 2$ is also the case in which the p -adic Hadamard gate does not exist. The proofs use p odd in several places (Hensel, $T = -S/2$ in the Newton step).

Q: *You randomise only one parameter in a two-block ansatz. What about the full parameter vector?*

Correct: this follows McClean et al.’s setting. Shallow-circuit and local-cost refinements in the style of Cerezo et al., and the behaviour of the full parameter vector, are open.

Key numbers: $\Pr[\text{maximal}] \geq 1 - \frac{1}{p} - 10p^{1-r}$ · witness $r = 2^{n-1}$ · $n = 10, p = 3$: correction $< 3^{-500}$
· $|\tilde{\mathbb{S}}_N| = p^{2N-1} - (-1)^N p^{N-1}$